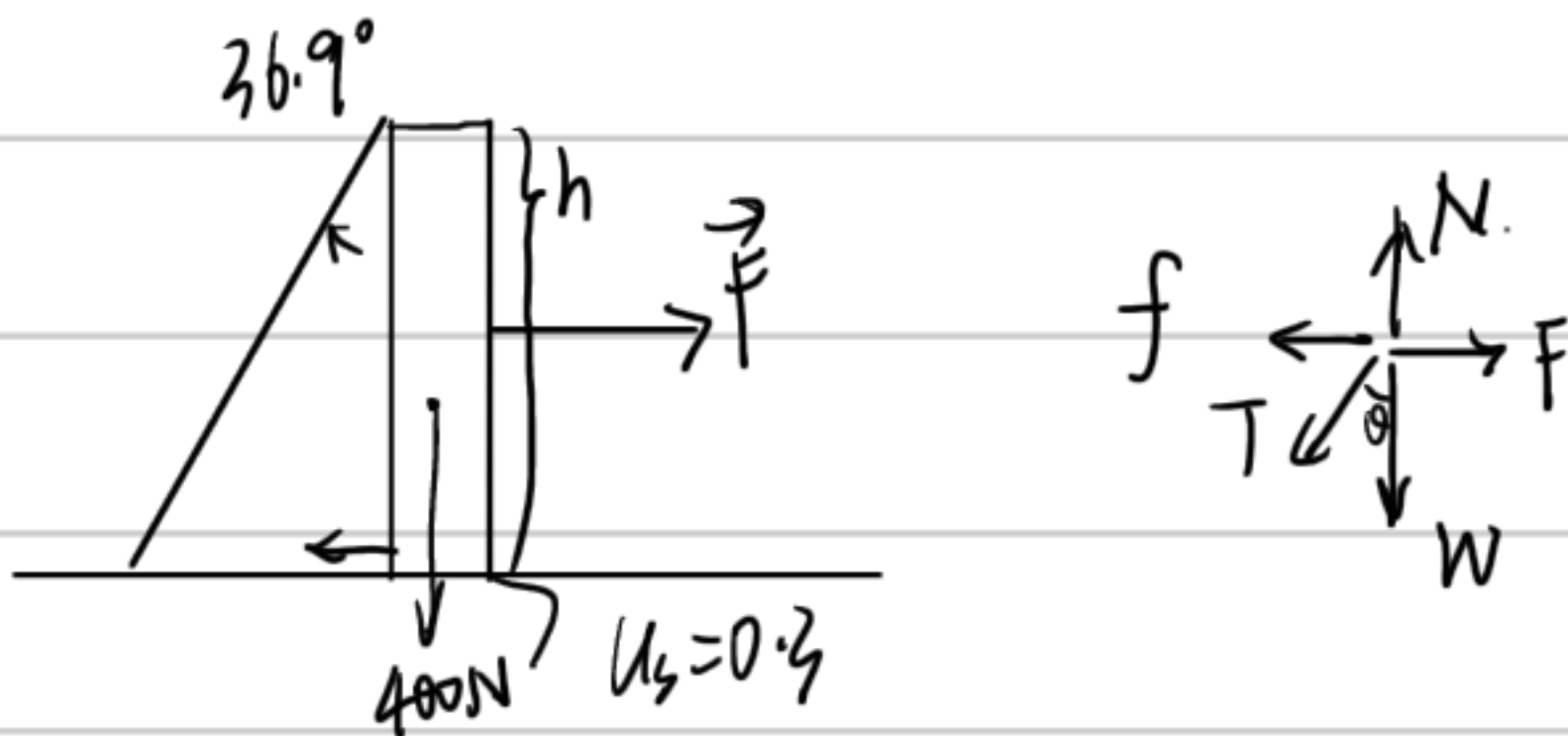


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P1. To make sure the handle end of the bat $v=0$, the rotational velocity and the translational velocity must cancel out.

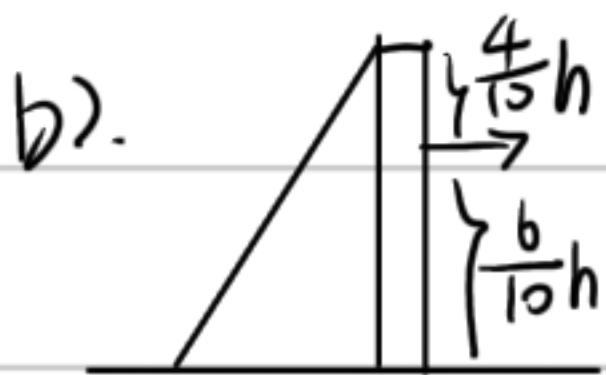
$$\begin{cases} J = mv \\ (1-d)J = I_{cm}\omega \\ v = d\omega \end{cases} \Rightarrow \lambda = \frac{I}{md} + d = 0.71 \text{ m}$$

P2.



take the upper end of the rope as pivot, when it's about to slip

$$\begin{cases} N = T \cos \theta + W \\ \mu_s N + T \sin \theta = F \\ \mu_s N h = F \cdot \frac{h}{2} \end{cases} \Rightarrow \mu_s \frac{F + \tan \theta W}{\mu_s + \tan \theta} = \frac{1}{2} F \quad F_1 = 400 \text{ N.}$$



$$\begin{cases} N = T \cos \theta + W \\ \mu_s N + T \sin \theta = F \\ \mu_s N h = F \cdot \frac{4}{10} h \end{cases} \Rightarrow F_2 = 750 \text{ N}$$

$$\mu_s \frac{F + \tan \theta W}{\mu_s + \tan \theta} = \frac{4}{10} F$$

$$\begin{cases} N = T \cos \theta + W \\ \mu_s N + T \sin \theta = F \\ \mu_s N h \geq F \cdot \frac{4}{10} h \end{cases}$$

$$\mu = \frac{0.3}{1.05} = 0.286$$

$$H_{critical} = (1 - \mu) h = 0.714 h$$

P3.

$$\begin{cases} \gamma = \frac{F/A}{\Delta L/L} \\ A = \pi \left(\frac{D}{2}\right)^2 \end{cases}$$

$$D = 2 \sqrt{\frac{FL}{\Delta L \pi \gamma}}$$

P4. $Y = \frac{W / A}{\Delta L / L_0}$ $F = k \Delta X$

$$K = \frac{W}{\Delta L} = \frac{AY}{L_0}$$

P5. $A_c = \frac{1}{2} A_D \Rightarrow V_c = 2 V_D$

$$\left\{ \begin{array}{l} P_{atm} + \rho g h_1 = P_c + \frac{1}{2} \rho V_c^2 \\ P_c = P_{in} \\ P_{in} + \rho g h_2 = P_{atm} \\ P_c + \frac{1}{2} \rho V_c^2 = P_{atm} + \frac{1}{2} \rho V_D^2 \end{array} \right.$$

$$P_c = P_{in}$$

$$P_{in} + \rho g h_2 = P_{atm}$$

$$P_c + \frac{1}{2} \rho V_c^2 = P_{atm} + \frac{1}{2} \rho V_D^2$$

$$h_2 = 3h_1$$

P6 (a) $\left\{ \begin{array}{l} P_{atm} + \rho g y = P_{hole} + \frac{1}{2} \rho V^2 \\ P_{hole} = P_{atm} \end{array} \right.$

$$V = \sqrt{2gy}$$

$$\left\{ \begin{array}{l} \frac{1}{2} g t^2 = h - y \\ D = Vt \end{array} \right.$$

$$D = \sqrt{4y(h-y)}$$

(b) since $D = 2\sqrt{y(h-y)}$

$$D(y) = \frac{h-2y}{\sqrt{y(h-y)}}$$

when $y = \frac{1}{2}h$ $D'(y) = 0$ so D reaches maximum h